



- 2 (a) State the principle of conservation of momentum.

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..... [2]

- (b) A ball X has mass 240 g and moves in a straight line on a horizontal frictionless surface with an initial speed of 16 m s^{-1} . The ball collides with a stationary ball Y that has mass 480 g. After the collision, ball X is stationary, as shown in Fig. 2.1.

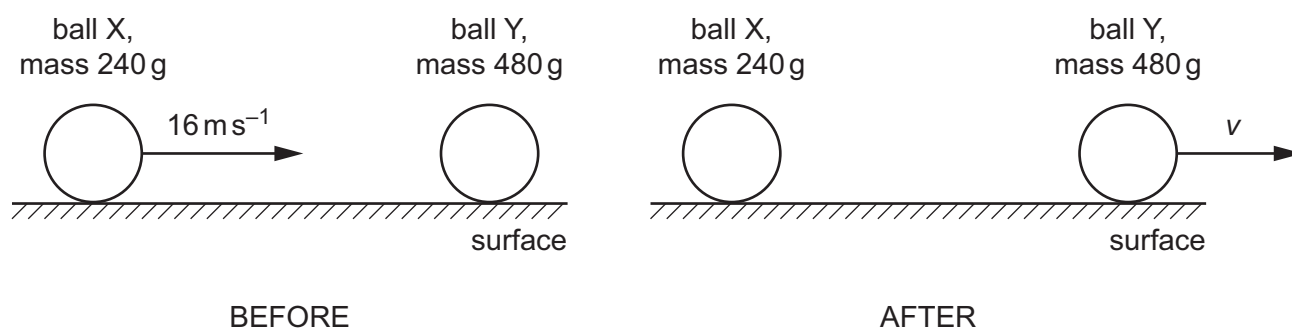


Fig. 2.1

- (i) Show that the speed v of ball Y after the collision is 8.0 m s^{-1} .

[1]

- (ii) Calculate the change in the total kinetic energy ΔE_K of the balls due to the collision.

$\Delta E_K = \dots\dots\dots \text{ J [3]}$





(c) The collision in (b) lasts for a time of 2.0 ms. Assume that the contact force between the balls is constant during this time.

(i) Determine the magnitude and direction of the force exerted on ball X by ball Y during the collision.

magnitude = N

direction [3]

(ii) Compare the magnitude and direction of the force exerted on ball Y by ball X during the collision with the answers in (c)(i). No further calculations are required.

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 [2]

[Total: 11]

